

Weekly Progress Report

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Domain: Core Java Development

Date of Submission: 20 June 2026

Week Ending: 01

I. Overview

During Week 1, the primary focus was on understanding the project requirements and designing the Banking Information System prototype in Core Java. I analyzed the functionalities required for the system, planned the application structure, and began implementing the core banking operations.

II. Achievements

1. Project Understanding and Planning

- Studied the Banking Information System requirements and identified the key modules.
- Designed the overall workflow of the application.
- Planned the structure for user accounts, transactions, and authentication.

2. User Registration Module

- Developed a user registration feature.
- Implemented functionality to collect user details such as name, address, contact information, and initial deposit amount.
- Generated unique account numbers for newly registered users.

3. Account Management Module

- Created functionality to view and update account information.
- Added confirmation messages for successful account updates.

4. Deposit and Withdrawal Module

- Implemented deposit and withdrawal functionality with balance validation.
- Updated account balances automatically after transactions.

5. Login and Security

- Designed a basic login system using username and password authentication.

III. Challenges

1. Account Number Generation

- Faced challenges in creating unique account numbers for each user.

2. Transaction Validation

- Implemented checks to prevent withdrawals exceeding available balances.

3. Data Organization

- Improved code structure by separating banking operations into different classes and methods.

IV. Lessons Learned

- Improved understanding of Classes, Objects, Constructors, and Encapsulation.
- Learned exception handling for invalid transactions and user input errors.
- Enhanced debugging and problem-solving skills.

V. Learning Resources

- Java Documentation
- Online Java Tutorials and YouTube Learning Resources

VI. Next Week's Goals

- Implement Fund Transfer Module.
- Develop Account Statement functionality.
- Add Data Persistence using file handling.
- Improve User Interface and validations.

VII. Additional Comments

This week provided valuable exposure to developing a real-world banking application using Core Java. I successfully implemented the foundational modules and gained practical experience in object-oriented programming and software development practices.